

Real-rooted h^* -polynomials of rank-two matroids with parallel classes of size at most three

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4 September 2026

Public Beta manuscript, version 0.1-beta. The mathematical proof and complete manuscript have passed project-internal zero-trust audits. External mathematical review and formal peer review are pending.

Abstract

Ferroni, Jochemko and Schröter proved that the Ehrhart h^* -polynomial of the base polytope of a rank-two sparse paving matroid is real-rooted. After loops are deleted, their rank-two hypothesis says that every parallel class has size at most two. We extend their result to parallel classes of size at most three. The new ingredient is a uniform estimate for the cubic correction term $p_{3,n}^*$ in their explicit h^* -formula. After the substitution $x = -\tan^2 \theta$, this estimate preserves the alternating signs used in their proof. We prove the estimate analytically for $n \geq 10$, settle $n = 9$ by an exact Sturm certificate, and check the finitely many smaller connected and disconnected cases exactly. Consequently the h^* -vector is log-concave and unimodal. We also identify the necessary Laguerre-polynomial inequality suggested by the scaling limit of the same method for an arbitrary fixed bound on parallel-class size.

1 Introduction

For a matroid M on a finite ground set, let $P(M)$ denote its base polytope and let $h^*(P(M), x)$ be the numerator of its Ehrhart series. Ferroni conjectured that the h^* -polynomial of every matroid base polytope is real-rooted [1]. Ferroni, Jochemko and Schröter subsequently proved this for rank-two sparse paving matroids [2]. They also gave an explicit formula for every rank-two matroid and proposed a precise analytic estimate as a possible route beyond the sparse paving case.

In a loopless rank-two matroid the rank-one flats are exactly the parallel classes. Sparse paving is therefore equivalent in this setting to requiring every parallel class to have size at most two. Our main result advances this boundary by one full structural layer.

Theorem 1. *Let M be a rank-two matroid. If every nonloop parallel class of M has size at most three, then $h^*(P(M), x)$ has only real zeros. Consequently its coefficient sequence is log-concave and unimodal.*

The proof is computer-assisted only at the finite boundary $n \leq 9$: the $n = 9$ estimate is certified by exact Sturm arithmetic, and the 31 connected types on at most eight nonloop elements and the six disconnected types are checked exactly. All larger ground sets are covered by one analytic argument.

The theorem does not settle the conjecture for all rank-two matroids. The largest parallel class is bounded by three, whereas a general rank-two matroid may have a parallel class of arbitrary size. Section 6 records the exact uniform inequality that remains for extending the method.

2 The rank-two formula

Deleting the loops of M removes coordinates that are identically zero on $P(M)$, so it preserves the lattice-isomorphism type of the base polytope. We may therefore assume that M is loopless and write n for the number of its remaining elements.

Suppose first that M is connected. Its parallel classes form a partition of the ground set, say with sizes $a_1, \dots, a_s$, where $s \geq 3$ and

$$a_1 + \dots + a_s = n.$$

Ferroni, Jochemko and Schröter [2, Proposition 5.1] proved

$$h^*(P(M), x) = \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} x^j - \sum_{i=1}^s p_{a_i, n}^*(x), \quad (2.1)$$

where

$$p_{a, n}^*(x) = \sum_{k=1}^a \sum_{j \geq 1} \binom{k}{j} \binom{n-k-1}{j-1} x^j. \quad (2.2)$$

They established [2, Lemma 5.3], for $a \in \{1, 2\}$, $n \geq 9$, and $0 < y \leq 1$,

$$\left| y^{n/2} p_{a, n}^* \left(\frac{y-1}{y} \right) \right| < \frac{a}{n}. \quad (2.3)$$

Our new ingredient is the next case.

Lemma 2. *For every $n \geq 9$ and $0 < y \leq 1$,*

$$\left| y^{n/2} p_{3, n}^* \left(\frac{y-1}{y} \right) \right| < \frac{3}{n}. \quad (2.4)$$

Directly from (2.2),

$$p_{3, n}^*(x) = 6x + (4n - 15)x^2 + \frac{(n-4)(n-5)}{2} x^3. \quad (2.5)$$

We prove Lemma 2.1 in the next two sections.

3 The uniform estimate for $n \geq 10$

Put

$$u = n \frac{1-y}{y} \geq 0, \quad y = \left(1 + \frac{u}{n} \right)^{-1}.$$

After substituting (2.5), inequality (2.4) is equivalent to

$$|P_n(u)| < 3 \left(1 + \frac{u}{n} \right)^{n/2}, \quad (3.1)$$

where

$$P_n(u) = -6u + \left(4 - \frac{15}{n}\right)u^2 - \frac{1}{2}\left(1 - \frac{9}{n} + \frac{20}{n^2}\right)u^3. \quad (3.2)$$

Let $T_{5,n}(u)$ be the Taylor polynomial of degree five at zero of $(1 + u/n)^{n/2}$. For $n \geq 10$, its sixth-order Taylor remainder is nonnegative on $u \geq 0$; this includes $n = 10$, when the remainder vanishes. Thus

$$\left(1 + \frac{u}{n}\right)^{n/2} \geq T_{5,n}(u). \quad (3.3)$$

It is therefore enough to prove $3T_{5,n}(u) - P_n(u) > 0$ and $3T_{5,n}(u) + P_n(u) > 0$.

For the upper inequality, direct expansion gives

$$3T_{5,n}(u) - P_n(u) = 3 + \frac{15}{2}u - A_n u^2 + B_n u^3 + E_n u^4 + F_n u^5, \quad (3.4)$$

where

$$A_n = \frac{29n - 114}{8n}, \quad B_n = \frac{3(n - 4)(3n - 14)}{16n^2},$$

$$E_n = \frac{(n - 6)(n - 4)(n - 2)}{128n^3}, \quad F_n = \frac{(n - 8)(n - 6)(n - 4)(n - 2)}{1280n^4}.$$

All four quantities are positive for $n \geq 10$, and

$$30B_n - A_n^2 = \frac{239n^2 - 2748n + 7164}{64n^2} > 0. \quad (3.5)$$

Indeed, on writing $n = 9 + v$, the numerator in (3.5) becomes $239v^2 + 1554v + 1791$. For $u > 0$, the arithmetic–geometric mean inequality and (3.5) yield

$$\frac{15}{2}u + B_n u^3 \geq 2\sqrt{\frac{15B_n}{2}}u^2 > A_n u^2.$$

Equation (3.4) is therefore positive; at $u = 0$ this is immediate.

For the lower inequality, write

$$3T_{5,n}(u) + P_n(u) = 3 - \frac{9}{2}u + u^2 R_n(u), \quad (3.6)$$

where

$$R_n(u) = C_n - D_n u + E_n u^2 + F_n u^3,$$

$$C_n = \frac{7(5n - 18)}{8n}, \quad D_n = \frac{(n - 4)(7n - 38)}{16n^2}. \quad (3.7)$$

The discriminant of the cubic $R_n - 1$ is

$$-\frac{(n-6)(n-4)^2(n-2)}{104857600n^{10}}Q(n), \quad (3.8)$$

and, with $n = 9 + v$,

$$Q(9+v) = 8703v^6 + 204112v^5 + 1743391v^4 + 6933784v^3 \\ + 13118701v^2 + 10391512v + 2399413.$$

Hence the discriminant in (3.8) is negative. The cubic $R_n - 1$ has positive leading coefficient and

$$(R_n - 1)(0) = \frac{27n - 126}{8n} > 0.$$

It has exactly one real zero, which must therefore be negative. Consequently

$$R_n(u) > 1 \quad (u \geq 0). \quad (3.9)$$

If $u \geq 37/10$, equations (3.6) and (3.9) give

$$3T_{5,n}(u) + P_n(u) > 3 - \frac{9}{2}u + u^2 > 0,$$

because the final quadratic is increasing on this interval and equals $1/25$ at $u = 37/10$.

If $0 \leq u \leq 37/10$, then $C_n \geq 21/8$, $D_n \leq 7/16$, and $E_n, F_n \geq 0$. Thus

$$3T_{5,n}(u) + P_n(u) \geq q(u) := 3 - \frac{9}{2}u + \frac{21}{8}u^2 - \frac{7}{16}u^3. \quad (3.10)$$

The two stationary points of q are $2 \pm 2\sqrt{7}/7$, where its values are $1 \mp \sqrt{7}/7$. Together with $q(0) = 3$ and $q(37/10) = 2009/16000$, this proves that q is positive on the entire interval. Equations (3.3)–(3.10) prove (3.1) for $n \geq 10$.

4 The finite boundary

For $n = 9$, let $D = (1 + u/9)^{9/2}$. Exact expansion gives

$$9D^2 - P_9(u)^2 = \frac{N(u)}{43046721}, \quad (4.1)$$

where

$$N(u) = u^9 + 81u^8 + 2916u^7 - 594864u^6 + 25627266u^5 \\ - 290698227u^4 + 1249949232u^3 - 1377495072u^2 \\ + 387420489u + 387420489.$$

The exact Sturm chain of N has degree profile

$$(9, 8, 6, 5, 4, 3, 2, 1, 0).$$

Its signs at zero and at positive infinity are respectively

$$(+, +, 0, -, -, +, -, -, -), \quad (+, +, +, -, -, -, +, +, -).$$

After zeros are omitted, both sequences have three sign variations. Hence N has no positive zero. Since $N(0) > 0$, equation (4.1) is positive for all $u \geq 0$, proving $|P_9(u)| < 3D$ and therefore Lemma 2.1 for $n = 9$.

For $3 \leq n \leq 8$, a connected loopless rank-two matroid is determined up to isomorphism by an integer partition of n into at least three parts. Restricting each part to size at most three leaves, for $n = 3, 4, 5, 6, 7, 8$, respectively

$$1, 2, 4, 6, 8, 10$$

possibilities. The accompanying exact verifier constructs (2.1) for all 31 partitions, factors off repeated factors, and uses Sturm root counts to place every zero, with multiplicity, in $(-\infty, 0)$. It also uses SymPy's exact algebraic-root routine as a second check within the same computer-algebra system.

5 Proof of the main theorem

Proof of Theorem 1. Assume first that M is connected and $n \geq 9$. Set

$$x = -\tan^2 \theta, \quad 0 \leq \theta < \frac{\pi}{2}.$$

The classical identity

$$\sum_j \binom{n}{2j} (-\tan^2 \theta)^j = \frac{\cos(n\theta)}{\cos^n \theta} \quad (5.1)$$

turns (2.1) into

$$h^*(P(M), -\tan^2 \theta) = \frac{q(\theta)}{\cos^n \theta}, \quad (5.2)$$

where

$$q(\theta) = \cos(n\theta) - \sum_{i=1}^s \cos^n \theta p_{a_i, n}^*(-\tan^2 \theta). \quad (5.3)$$

Put $y = \cos^2 \theta$. For $a_i = 1, 2$, use (2.3), and for $a_i = 3$, use Lemma 2.1. Since the a_i sum to n , equations (5.3) and the strict estimates give

$$|q(\theta) - \cos(n\theta)| < \sum_{i=1}^s \frac{a_i}{n} = 1. \quad (5.4)$$

At the points $\theta_j = j\pi/n$, equation (5.4) makes $q(\theta_j)$ strictly positive for even j and strictly negative for odd j .

When n is odd, all points with $0 \leq j \leq \lfloor n/2 \rfloor$ lie below $\pi/2$. When $n = 2m$ is even, the last point is the endpoint $\pi/2$, outside the domain of (5.4). In this case every correction in (5.3) is a linear combination of terms

$$(-1)^j \sin^{2j} \theta \cos^{n-2j} \theta, \quad 1 \leq j \leq 3.$$

Since $n \geq 10$, the exponent $n - 2j$ is at least four. All corrections therefore tend to zero at the endpoint, and q extends continuously with

$$q(\pi/2) = \cos(n\pi/2) = (-1)^m.$$

Thus, in both parity cases, q has a zero in each of the $\lfloor n/2 \rfloor$ consecutive open intervals between the sign-test points. Under the injective map $\theta \mapsto -\tan^2 \theta$, these become distinct negative zeros of $h^*(P(M), x)$.

The first sum in (2.1) has degree $\lfloor n/2 \rfloor$, while every correction has degree at most three. For $n \geq 9$, the former degree is at least four, so its positive leading term cannot be cancelled. Hence the number of zeros found equals the degree, and all zeros are real. The smaller connected cases were settled in Section 4.

It remains to treat disconnected matroids. After loops are deleted, a disconnected rank-two matroid is $U_{1,a} \oplus U_{1,b}$, where $a, b \geq 1$. Under our hypothesis $a, b \leq 3$, and $P(M)$ is the product $\Delta_{a-1} \times \Delta_{b-1}$. For the six unordered pairs

$$(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3),$$

the corresponding h^* -polynomials are

$$1, \quad 1, \quad 1, \quad 1+x, \quad 1+2x, \quad 1+4x+x^2.$$

They are all real-rooted. This completes the proof of Theorem 1.1. The log-concavity and unimodality conclusions follow from Newton's inequalities and the nonnegativity of Ehrhart h^* -coefficients. $\square$

6 The boundary of the method

Ferroni, Jochemko and Schröter [2, Section 6.1] suggested proving, for every fixed a , that for all sufficiently large n ,

$$\left| y^{n/2} p_{a,n}^* \left(\frac{y-1}{y} \right) \right| < \frac{a}{n}. \quad (6.1)$$

The same sign-interlacing argument would then cover rank-two matroids whose parallel classes have size at most a , subject to the degree condition. A useful necessary asymptotic test is visible in the scaling used above. For fixed a , the hockey-stick identity gives

$$\lim_{n \rightarrow \infty} n p_{a,n}^*(-u/n) = \sum_{j=1}^a (-1)^j \binom{a+1}{j+1} \frac{u^j}{(j-1)!} = -u L_{a-1}^{(2)}(u), \quad (6.2)$$

where $L_m^{(2)}$ is a generalized Laguerre polynomial. Since $(1+u/n)^{-n/2} \rightarrow e^{-u/2}$, pointwise passage to the limit in (6.1) would require the non-strict inequality

$$\sup_{u \geq 0} |u e^{-u/2} L_{a-1}^{(2)}(u)| \leq a. \quad (6.3)$$

Numerical evidence suggests the stronger strict inequality, which would give a useful pointwise margin. Even that strict bound would not by itself imply the finite- n estimate (6.1) on the unbounded interval $u \geq 0$: a uniform convergence argument and separate tail control would still be needed. Standard Laguerre bounds are too weak after multiplication by u , and treating $a = 4, 5, \dots$ by separate increasingly high-degree polynomial certificates would not supply a uniform mechanism. The present note therefore stops at the natural $a = 3$ boundary.

7 Verification and reproducibility

The supplementary Python script uses integer and rational arithmetic through SymPy. It performs the following checks:

1. derives (2.5) directly from (2.2);
2. reconstructs both Taylor expansions in Section 3 and verifies their coefficient, discriminant, stationary-point and endpoint certificates;
3. reconstructs $N(u)$ in (4.1), its Sturm profile and endpoint signs;
4. includes a positive-root negative control for the Sturm routine;
5. checks all 31 small connected partitions using exact Sturm counts and an additional exact algebraic-root check in the same SymPy stack;
6. derives and checks all six disconnected cases.

All failure conditions are explicit exceptions, not Python `assert` statements, so the checks remain active under `python -O`. The script is a reproducibility aid and supplies the exact finite portion of the proof; it does not replace the analytic argument for $n \geq 10$.

The audited environment used Python 3.13.11 and SymPy 1.14.0. The SHA-256 of the audited script is 00FA0B2A2A7481B6B7E562EDC5EFF2DD4DD63ED776F3BA1E39FCFCED7EFEF9C9.

8 Prior-work and novelty boundary

Ferroni, Jochemko and Schröter [2] supplied both the general rank-two formula and the sign-interlacing method, and proved the case in which parallel classes have size at most two. They explicitly identified (6.1) as a possible route to any fixed upper bound. Our contribution is limited to proving the next structural case $a = 3$, including all finite and disconnected boundaries, and to identifying the Laguerre scaling limit and the additional uniformity obligations for a parameter-uniform extension.

Jiang [3] later gave combinatorial formulas for the h^* -polynomials of all positroid polytopes. Since every rank-two matroid is isomorphic to a positroid [2, Corollary 6.2], that work gives a broader formula-level context, but it does not state the real-rootedness theorem proved here.

Targeted searches completed on 4 September 2026 found no paper directly covering parallel classes of size at most three. This supports candidate novelty but does not establish global priority. We do not claim to settle real-rootedness for all rank-two matroids or for all matroid base polytopes.

AI-assisted research disclosure

Carptopus is the responsible author and takes responsibility for the claims, proofs, source use and released artifacts. OpenAI Codex was used for AI-assisted literature organization, proof development and stress testing, exact verification, adversarial auditing and manuscript preparation.

References

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